Does scrolling more mean feeling worse?

Executive Summary, ACDT Project 1: Revisit “Tackling the Conventional Wisdom”
GROUP32: Chae Seungrok, Im Junho, Jaemin Ok, Jeongwoo Park, Kim Dohyun, Park Jihu
Website: dhkim060225-web.github.io/group32-sns-mental-health
Code: github.com/dhkim060225-web/group32-sns-mental-health

The conventional wisdom and our question

Many people believe that the more time students spend on social media, the worse they feel and the worse they do in class. We set out to test whether passive social media use lowers grades. The only usable dataset had no GPA and no measure of passive versus active use, so we asked: **is more daily social media use associated with a lower self-rated mental health score, and how much of that link survives other explanations?**

Data and approach

We used the *Students' Social Media Addiction* dataset (Kaggle): 705 self-reported survey responses, 13 variables. After removing one duplicated row, **n = 704** students from **110** countries remained. We fitted a simple linear regression with a t-test and 95% confidence interval, validated it on 200 random 80/20 train/test splits, and then tested competing explanations: country clustering, platform type, sleep as a pathway, and double machine learning (DML) to remove measured background differences.

Key findings

Finding	Evidence
Heavier use goes with lower mental health	−0.70 points per extra hour (95% CI −0.74 to −0.66; t = −35.5, p < 0.001); R^2 = 0.64
The model predicts unseen students as well as seen ones	Test R^2 0.64 on average across 200 splits (range 0.48–0.73); error 0.66 vs 1.10 for guessing the mean. For one student the 95% prediction interval is about ±1.3 points.
Country explains more than hours do	Country alone explains 69% of score variation (63% adjusted). Within countries the slope is −0.58; the country-clustered CI is −0.80 to −0.61.
Platform type changes the relationship	Slopes range from −0.41 (WhatsApp) to −1.05 (Twitter); one common slope is rejected, F(8, 650) = 3.59, p < 0.001.
Sleep carries part of the link	About 20% runs through sleep (indirect effect −0.14, 95% CI −0.18 to −0.10).
The link survives measured background	After DML removes age, gender, academic level, relationship status and country: −0.63 per hour (95% CI −0.68 to −0.59).

What the data actually shows

The conventional wisdom points in the right direction but tells the wrong story. Hours matter, roughly 0.6 points per hour even after removing measured background, but they are tangled up with where students live, how much they sleep and which platform they use. Because the data is cross-sectional and self-reported, it cannot tell us whether heavy use lowers mood or low mood drives heavy use, and unmeasured stress could drive both.

Limitations

The sampling frame is undocumented, so results describe this dataset rather than all students. The data looks unusually clean: the addiction score correlates −0.95 with mental health, every user of three platforms gives the same score, and “social media hurts my studies” answers jump from 0% to 100% across usage bands. Students are clustered by country, residuals are not normal, and stress and purpose of use were never measured. We treat the results as a strong teaching example, not as evidence for policy.

Recommendations

- **Counsellors:** use heavy use as a reason to start a conversation, not as a diagnosis; ask about sleep first.
- **Students:** notice what the hours are for; a group chat and an endless feed may not affect you the same way.
- **Researchers:** measure stress and purpose of use, follow the same students over time, and sample within countries.